Handout notes for HDF 100

FLAX

Introduction

Flax fibre is another natural cellulosic fibre derived from the stem of fibres. Although its source is different from that of cotton, its fibre and fabric properties are similar to those of cotton.

Growth and production

Flax is a bast fibre – a woody fibre obtained from phloem of plants. The term linen refers to cloth made from flax. The flax plant requires a temperate climate with generally cloudy skies and adequate moisture. Bright sunlight and high temperatures are damaging unless alternated with abundant rainfall.

The plant grows to a height of 3 – 4 feet and the flax for fibre is pulled before seeds are ripe. Major producers of flax include USSR, Poland, Belgium, Ireland, France, Newzealand, and Netherlands.

Processing of flax

1. Pulling and rippling

Flax fibre may be pulled by hand or mechanical pullers. It is important to keep the roots intact as fibres extend below the ground surface. After drying, the plant is rippled, i.e. it is pulled through special threshing machines that remove the seed bolls on pods.

2. Retting

To obtain fibres from the stalk, the outer wood portion must be rotted away. This process, known as retting can be accomplished by any of the several procedures.

(a) **Dew retting –** Involves the spreading of flax on the ground, where it is exposed to the action of dew and sunlight. This natural method of retting gives uneven result but provides the strongest and most durable linen. It requires a period of 4 to 6 weeks.

(b) **Pool retting –** Is a process whereby flax is packed in sheaves and immersed in pools of stagnant water. Bacteria in the water rot away the outer stalk covering. The time required is 2 to 4 weeks.

(c) **Tank retting** – Similar to pool retting but utilizes large tanks in which the flax is stacked. The tanks are filled with water, which increases the speed of bacterial action. The tank retting requires only a few days. Both pool and tank retting give good quality flax which is uniform in strength and light in colour.

(d) **Chemical retting or steam retting** – Is accomplished by stacking the flax in tanks, filling the tanks with water, and adding chemicals such as sodium hydroxide, sodium carbonate, or dilute sulphuric acid. Chemical retting can be completed in a matter of hours instead of days or weeks. However, it must be carefully controlled in order to prevent damage to the fibre.

3. **Breaking and scutching**

After the retting is complete the fibre is rinsed and dried. The stalks are then bundled together and passed between fluted rollers that break the outer woody covering into small particles. It is then scutched to separate the outer covering from the usable fibre.

4. **Hackling**

After scutching the flax fibres are hackled or combed. This separates the short fibres called tow from long fibres called line. This is accomplished by drawing the fibres between several sets of pins, each successive set finer than the preceding set and is similar to the combing operation used for cotton. It prepares the flax fibres for the final steps yarn manufacture. As the fibres are removed from the hackling machine they form a sliver.

5. **Spinning**

The flax sliver is drawn into yarn, and twist is imparted. Flax fibres are spun either dry or wet, but wet spinning is considered to give the best quality yarn. The final yarn processing is similar to that used for cotton fibre.

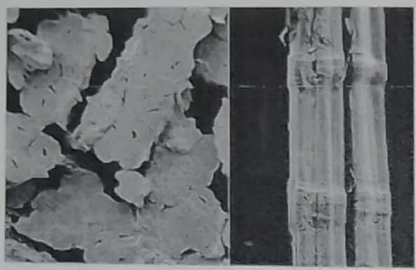
Fibre Properties

Microscopic properties

Flax fibre is composed of buddles of fibrils of fibre cells held together by a bonding or gummy substance. Under the microscope the longitudinal view of the fibre shows the width to be quite irregular. The central canal or lumen casts a shadow, giving a slightly darker effect down the centre. The points at which the fibre width changes are marked by swellings and irregular joint formation called nodes. These are similar to the joints in bamboo.

The cross-section view of flax clearly shows the lumen the thick outer wall and somewhat polygonal shape. Immature fibres may be oval in shape and have larger lumen than mature fibres.

Cross-sectional view **Longitudinal view**



Source: swicofil.com

Physical properties

1. Shape and appearance

Flax fibre is not as fine as cotton. Flax cells have an average diameter of 15 to 18 microns, and vary in length from 0.63 – 6.35cm. Bundles form the actual fibres used in spinning into yarns and these bundles may be 42.7 – 50cm long. Line fibres are usually more than 30.5cm long while tow fibres are shorter. The natural colour of flax varies from light ivory to gray.

2. **Lustre**

Flax fibres have a high natural lusture with an attractive sheen.

3. **Tenacity (strength)**

Flax is a strong fibre. The normal tenacity ranges between 5.5. grams per denier. Fabrics of flax are durable and easy to maintain because of the fibre strength. When wet, the fibre is about 20% stronger than when dry. Most linen fabrics for apparel use are given various resin finishes which reduce strength. Flax has low pliability (flexibility) and this may reduce serviceability where frequent bending is required in use.

4. **Elastic recovery and elongation**

The amount of elongation that flax will undergo before breaking is very small. When dry, the fibre will extend only 2.7 to 3.3%. It has very little elasticity.

5. **Resiliency**

Linen fabrics crease and wrinkle badly. They are stiff and possess little resiliency. However, finishes can be applied to offset this to some degree.

6. **Density**

The density of flax is $1.5 g/cm^3$ and in comparable to other cellulosic fibres.

7. **Moisture regain**

Flax has a standard moisture regain about 12%. The saturation regain is comparable to other cellulosic fibres. Flax has outstanding wicking properties, which makes it possible to move moisture along the fibres and yarns as well and absorbs moisture.

8. **Dimensional stability**

Flax fibres do not stretch or shrink to marked degree. However, yarns and fabrics are subject to some relaxation shrinkage unless pre-shrunk. Ironing fabrics while damp help stretch them back to their original size.

Thermal properties

Flax burns quickly. It is highly resistant to decomposition by dry heat and will withstand temperatures up to 150° C for long periods with little change in properties. Prolonged exposure above 150° C will result in gradual discolouration. Fabrics may be ironed safely with temperatures up to 260° C.

Chemical properties

Flax is highly resistant to alkaline solution and dilute acids._Concentrated acids and dilute hot acids cause deterioration._Flax has excellent resistance to dry cleaning solutions and other organic compounds used for normal maintenance. It however gradually looses strength when fabrics are exposed to sunlight but this is not serious. Flax fabrics make good choice for curtains. If properly stored, linen will age well.

Biological properties

Dry linen has excellent resistance to mildew, but if the fabric is moist mildew will grow rapidly and damage the fibre. Flax has good resistance to most household insects and pests. However silverfish may damage the fibre if it is starched.

Use and Care of Linen Fabrics

Flax makes a wide variety of fabrics from sheer and loose to heavy and compact. Linen is frequent choice for table coverings because it wears well and is extremely attractive. The natural resistance of flax to chemicals provides a fabric that is easily maintained. The linen fabrics have a long life. Linen fabrics are easily laundered and it's the preferred method. The fabrics do not require bleaching unless stained.

SILK

Introduction

Silk is a natural protein fiber and the only natural fiber that comes in a filamentous form. It is known as a luxurious fiber because of the beautyof the textile products it produces. The fabrics made from silk are expensive due to the expensive growth and production procedure.

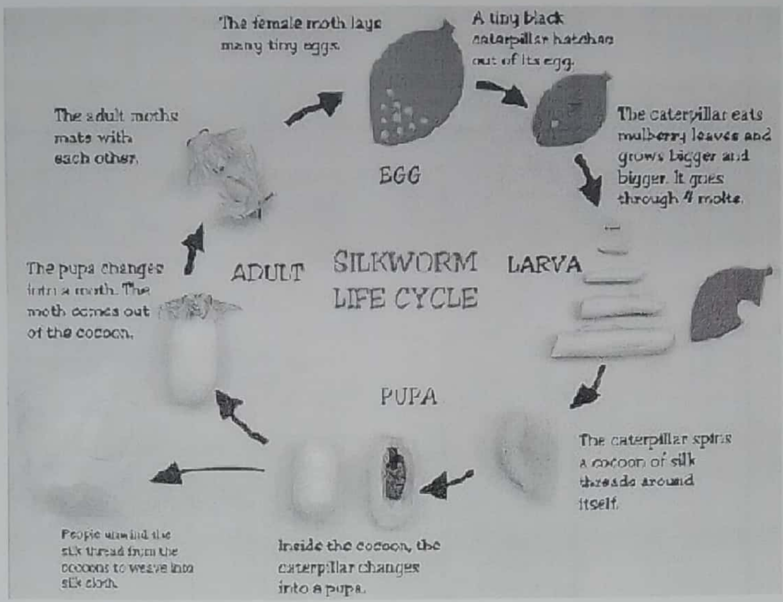
Growth and production

Female *Bombyx mori* lays as many as 700 eggs. The eggs are incubated and hatch out within 10 days. The new larvae is about ¼" (0.64cm) long. During the growth cycle, of the larva, it sheds its outer skin (molts) four times. Each time, it grows a new skin and increases in size. After the fourth molting, it eats for 10 more days, making a total eating period of about 28 days. The caterpillar increases about 10,000 times the weight of the newly hatched larvae, 3inches in length and ¼ to ½ inches in diameter. Its now ready to spin its cocoon, or chrysalis case. The larvae attaches itself to a specially constructed straw. From the rears its head, it begins to spew the silk liquid, which harden upon contact with air. The larvae pins by moving its head in a figure eight motion and constructs the cocoon from the outside. As it spins the larva decreases in size and upon completion of the cocoon the caterpillar changes into the dormant chrysalis stage (pupa).

Cocoons can be stored for long periods by killing the chrysalis using heat.

This is called stifling. Cocoons not stifled are permitted to hatch and the moths emerge breed and lay a new batch of eggs so that the cycle can start anew. Silk worms extrude the liquid from two tiny orifices or spinnerets in its head. The filaments, coated with a gummy substance, called sericin, comes from two glands within the worm. In addition to the sericulture of *Bombyx mori* several other silk producing moths are *Antheraea myllita* and *Antheraea Penny*, which produce tussah (wild) silk. These wild insects feed on oak leaves and their cocoons are collected from trees and shrubs. The larva of these species grows much larger than cultivated silk worm up to (15cm) 6". Major producers of silk are Japan, Thailand, and China

The life cycle of a silkworm



Source: www.sites.google.com

Processing of silk fiber

Reeling

Silk filaments are unwound from the cocoon in a manufacturing plant called a filature. Cocoons are placed in hot water to soften the gum and the surfaces are brushed to find the ends of the filaments. The ends are threaded through a guide and wound onto a wheel called reel hence the name reeling. This requires skilled operators and sophisticated machinery.

Throwing

As the fibres are combined and pulled onto the reel twists can be inserted to hold the filaments together. This is called throwing resulting in thrown yarn. Thrown yarn differs in the amount of twists and the methods of combining single yarns e.g.

(a) **Tram silk** – is a low twist ply yarn formed by combining two or three single strands. It's moderately strong and is frequently used for fillings yarns.

(b) **Organzine** – Is a two or more ply yarn with a medium twist. It's very strong and is used for warp yarns.

(c) **Crepe organzine** – Is a very highly twisted yarn used in crepe fabrics and chiffons.

(d) **Singles** – Is a strand of silk fiber held by low, medium, or high twist. A single yarn can be used both for warp or filling and in knitting.

(e) **Grenadine** – is tightly twisted ply yarn composed of two or three singles. The ply twist of each single strand.

Degumming

Sericin remains on the fibre during reeling and throwing and frequently throughout the fabric construction procedures.

Before the final finishing, the gum is removed by boiling the fabric in soap and water. If stiffness is desired in the completed fabric some of the sericin may be replaced but it is not usually desirable coz its presence increases the tendency for silk to water-spot.

Fiber Properties

<u>Microscopic Properties</u>

- Cultivated degummed silk, viewed longitudinally under a microscope resembles a smooth transparent rod.
- Silk with sericin on it has a rough irregular surface.
- Cross-sectional views of silk show triangular fibres with no inner markings.
- Two filaments usually lie with their flat sides together.
- The two face to face filaments are called brins.

Cross-Sectional View **Longitudinal View**

b) cross section of silk fibre *a) longitudinal view of silk fibre*

Source: www.cci-icc.gc.ca

Physical properties

(a) **Shape and Appearance**

- Silk filaments are very fine and long.
- They frequently measure about 915 – 1190 meters and can be as long as 2750m.
- The width of silk is from 9 – 11 microns.
- The fibres are smooth with high natural lusture and sheen. They are off-white to cream in colour. Occasionally they can be yellow or light brown.
- Wild silk is uneven, has slightly less lusture and is tan to medium brown in colour.

(b) **Lustre**

- Cultivated silk is a lustrous fibre.
- Wild silk is somewhat dull.

(c) **Tenacity**

- Silk is one of the strongest natural fibres used in textile products
- It has tenacity of 2.45-5.1g/ when dry.
- Wet strength is about 80-85 of the dry strength.

(d) **Elastic recovery and elongation**

- Silk has good elasticity and elongation.
- When dry the elongation is 10-25% while when wet it will elongate as much as 33 – 35%.
- At 2% elongation the fibre has 92% elastic recovery.

(e) **Resiliency**

Silk has medium resiliency. Creases will hang out relatively well, but not so quickly or completely as for wool.

(f) **Density**

Density (or specific gravity) of silk is cited as 1.25 – 1.34 g/cm^3.

(g) **Moisture Regain**

- Silk has relatively high standard moisture regain of 11%.

- At saturation, the regain is 25 – 35 %.
- The relatively high absorption is helpful in applying dyes and finishes to silk. However unlike other fibres silk also absorbs impurities such as metal Salts. These tend to cause damage to silk by weakening the fibre.

(h) **Dimensionally Stability**

- Silk fabrics have good resistance to stretch or shrinkage when laundered or dry-cleaned.
- Crepe fabrics will shrink when wet but can be steered back to shape.

Thermal properties

- Silk will ignite and continue burning when there is a source of flame. After removal from the source it will sputter and eventually extinguish itself.
- It leaves a crisp brittle ash and gives off a smell like that of burning hair of feathers. Silk burns in a similar manner to wool.
- Silk can withstand temperatures of up to 135° C for long periods of time.
- However, temperatures up to 177° C cause rapid degradation.
- If ironed above 150° C it easily scorches and white silk will turn yellow if pressed with an iron hotter than that.
- Like other protein fibres, silk fabrics tend to be warmer than fabricsof cellulosic fibres.
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Chemical properties

(a) **Effect of Akalis**

- Silk is damaged by strong alkalis and will dissolve in heated caustic soda (NaOH) (reacts more slowly than wool).
- Weak alkalis like soap, borax and ammonia cause little or no damage unless if in contact for a long time.

(b) **Acids**

- Silk can be decomposed by strong mineral acids.
- Medium concentrations of Hydrochloric acid (HCL) will dissolve it and moderate concentrations of other mineral acids cause fibre contraction and shrinkage.

(c) **Effect of Organic Solvents**

- Cleaning solvents and spot removers do not damage silk fibres.

(d) **Sunlight, Age and Miscellaneous**

- Sunlight tends to accelerate silk decomposition
- Silk requires careful handling and adequate protection to withstand ravages of time.
- Oxygen causes gradual decomposition hence silk should be stored in sealed containers.

(e) **Static Charge**

- Silk is a poor conductor of electricity hence it builds up static charge.

Biological Properties

- Silk is resistant to attack by mildew and relatively reacts to other bacteria and fungi.
- It is resistant to clothes moth, but carpet beetles will eat it.

Use and Care of Silk

- It has and still is used for luxury fabrics and high fashioned items. It is frequently considered a luxurious fabric because of its smooth and soft feel, or hand.
- Dry cleaning is the preferred method for care of silk fabrics and products.
- It can be carefully laundered in mild synthetic soap and warm water, with minimal handing.
- Should be thoroughly rinsed and water extracted using a towel in which you roll the garment and hang in a cool place.
- Silk should be ironed or pressed at medium to low temperatures.
- Silk is versatile and can be used in variety of fabric for home furnishings.
- Many silk fabrics cost considerably more than similar fabrics of man-made fibres.
- Its use is limited primarily to apparel and home furnishings.